Family-level Abundance Matrix Construction (Kraken2 vs KrakenUniq)

The bash and R code in this report were written by me, with assistance from the Claude AI on specific steps: to confirm a methodological choice or to apply a correction/debug.
Tools: Kraken2 + GTDB; KrakenUniq + MicrobialNT; R 4.5.2 (vegan 2.7-3, dplyr, RColorBrewer).

Prepare the input tables for downstream comparison plots at family level between Kraken2 and KrakenUniq.
Two *samples×families* matrices were produced on Dardel

Step 1: Locate the report files

```
ls /cfs/klemming/projects/supr/archaeogenetics/mm1202/Kraken2_results/*.report | wc -l
ls /cfs/klemming/projects/supr/archaeogenetics/mm1202/KrakenUniq_results/*.report | wc -l
```

- 250 Kraken2 reports
- 251 KrakenUniq reports: one with anomalous name KrakenUniq_rep_.merged.fq.gz.report (no sample ID), skipped downstream
- Sample IDs identical between the two tools

Step 2: Single-sample test (Kraken2)

In Kraken2 reports the rank label is in column 6, with F for family.

```
awk '$6=="F" {print $1"\t"$NF}' Kraken2_rep_P32015_1001_S28_L005_fastp_merged.fq.gz.report | sort -k1 -rn | head -20
```

Output: 2 columns sorted by % decreasing.
Top hit: *Burkholderiaceae* 0.47%.

Step 3: Loop over all Kraken2

```
mkdir -p /cfs/klemming/…/Kraken2_analysis/per_sample # Claude's help
cd /cfs/klemming/.../Kraken2_results
for f in Kraken2_rep_*.report; do
   sample=${f#Kraken2_rep_}
   sample=${sample%_fastp_merged.fq.gz.report}
   awk '$6=="F" {print $1"\t"$NF}' "$f" | sort -k1 -rn \
    > ../Kraken2_analysis/per_sample/${sample}_families.tsv
done
```

Output: 250 per-sample files in Kraken2_analysis/per_sample/.

Step 4: Single-sample test (KrakenUniq)

KrakenUniq has a different format: header lines start with #, the rank label is in column 8 with the literal word family, and tax names may contain spaces.

```
awk '!/^#/ && $8=="family" {name=$9; for(i=10;i<=NF;i++) name=name" "$i; print $1"\t"name}' \
KrakenUniq_rep_P32015_1001_S28_L005_fastp_merged.fq.gz.report \| sort -k1 -rn | head -20
```

Output: top hit Hominidae 42.94%.
Homo sapiens was excluded at species level in the original analysis, so the Hominidae family is still retained: relevant because Kraken2/GTDB does not include eukaryotes, so this family will need handling at the comparison stage.

Step 5: Loop over all KrakenUniq

Anomalous files (sample ID not starting with P<digit>) are skipped:

```
# Claude's help
mkdir -p /cfs/klemming/.../KrakenUniq_analysis/per_sample
cd /cfs/klemming/.../KrakenUniq_results
for f in KrakenUniq_rep_*.report; do
   sample=${f#KrakenUniq_rep_}
   sample=${sample%_fastp_merged.fq.gz.report}
   [[ ! "$sample" =~ ^P[0-9] ]] && continue
   awk '!/^#/ && $8=="family" {name=$9; for(i=10;i<=NF;i++) name=name" "$i; print $1"\t"name}' "$f" \
    | sort -k1 -rn > ../KrakenUniq_analysis/per_sample/${sample}_families.tsv
done
```

Output: 250 per-sample files (1 anomalous file skipped).

Step 6: Union of all families per tool

```
cut -f2 Kraken2_analysis/per_sample/*_families.tsv    | sort -u > Kraken2_analysis/all_families.txt
cut -f2 KrakenUniq_analysis/per_sample/*_families.tsv | sort -u > KrakenUniq_analysis/all_families.txt
```

- **Kraken2**: 5460 unique families. Many are GTDB alphanumeric placeholders (e.g. 0-14-0-10-38-17, 01-FULL-45-15b) for taxa without an official name.
- **KrakenUniq**: 1662 unique families, all with standard names; includes eukaryotes (e.g. Acanthamoebidae, Acanthocerataceae).

Step 7: Build the *samples×families* matrix

Same one-liner applied to both tools (only paths change):

```
# Claude's help
awk -v famfile=<all_families.txt> '
  BEGIN{ FS="\t"; while((getline l<famfile)>0){F[++n]=l; ix[l]=n}; close(famfile);
        printf "Sample"; for(i=1;i<=n;i++) printf "\t%s",F[i]; print "" }
  FNR==1{ if(cur){ printf "%s",cur; for(i=1;i<=n;i++) printf "\t%s",(i in M?M[i]:"0"); print ""; delete M };
         cur=FILENAME; sub(/.*\/,"",cur); sub(/_families.tsv$/,"",cur) }
  { if($2 in ix) M[ix[$2]]=$1 }
  END{ if(cur){ printf "%s",cur; for(i=1;i<=n;i++) printf "\t%s",(i in M?M[i]:"0"); print "" } }
' <per_sample>/*_families.tsv > <output>.tsv
```

Each matrix has the sample name as first column, all families as header row, and the percentage from column 1 of the original report as cell value (0 if the family is absent in that sample).

Quality control: empty per-sample files

```
find <per_sample> -name "*_families.tsv" -size 0
```

- **Kraken2**: 1 empty file → P35761_1083_S83_L001. Inspecting the original report shows only one line: 100.00 2 2 0 0 U 0 unclassified (2 reads, all unclassified — failed sample).
- **KrakenUniq**: 3 empty files → P35761_1101_S101_L001, P34558_1098_S253_L005, P34558_1100_S255_L005.

These samples carry no family-level data and were excluded from their respective matrices.

Two samples × families matrices produced on Dardel

Tool	Database	Rows (samples)	Columns (families)	Excluded
Kraken2	GTDB (bacteria/archaea)	249	5460	P35761_1083
KrakenUniq	MicrobialNT (incl. eukaryotes)	247	1662	P35761_1101 P34558_1098 P34558_1100

Output files:

- /cfs/klemming/projects/supr/archaeogenetics/mm1202/Kraken2_analysis/kraken2_family_matrix.tsv
- /cfs/klemming/projects/supr/archaeogenetics/mm1202/KrakenUniq_analysis/krakenuniq_family_matrix.tsv

Step 8: Local Export and R Environment Setup

The two matrices were transferred from Dardel to the local machine via scp. An R script final_family_analysis.R was developed using the vegan, dplyr and RColorBrewer libraries.

Step 9: Data Cleaning and Filtering (R)

Before the comparison, the data underwent a cleaning process in R (final_family_analysis.R) to ensure the two datasets were ecologically comparable:

- the analysis was restricted to the **246 samples present in both tools** (the intersection of the two sample sets), instead of using the union of all 250 samples. This avoids zero-padding artifacts that would otherwise bias the multivariate analysis.
- since Kraken2/GTDB does not include eukaryotes, the **Hominidae family (human DNA) was removed from both datasets**. This prevented human contamination from masking the microbial signal, which accounted for ~43% of classified reads in KrakenUniq.
- one additional sample (P34558_1102_S257_L005) was excluded because, after removing Hominidae from KrakenUniq, it had zero microbial reads assigned at family level; biologically a sample dominated by human DNA with negligible microbial biomass, but unusable for the cross-tool comparison. Final number of samples in the analysis: **245**.
- for the Bray-Curtis dissimilarity step, each sample's vector was normalized so that **rowSum = 1**. It removes the effect of different total classified reads between samples and between tools, so that every sample contributes equally to the multivariate analysis.
- the comparative analysis was restricted to families sharing the exact same nomenclature between GTDB and MicrobialNT, identifying **362 shared families**.

Step 10: Statistical Analysis

Two levels of comparison were performed:

Taxonomic Correlation (Spearman)

Calculated for the 362 shared families to evaluate agreement on specific abundances.

Results:
- of the 362 shared families, only **201 had quantitative data in both tools**, the remaining 161 were nominally shared but had zero abundance in one tool across all samples
- among the 201 valid families, **the median Spearman rho was 0.28** (mean 0.31), with 31.8% above 0.5 and 9.5% above 0.8.

Strong agreement is concentrated on the dominant taxa:
- Burkholderiaceae (0.99)
- Gemmatimonadaceae (0.96)
- Rhodanobacteraceae (0.94)
- Steroidobacteraceae (0.94)
- Micromonosporaceae (0.94)

Global Community Structure (PCoA & Procrustes)

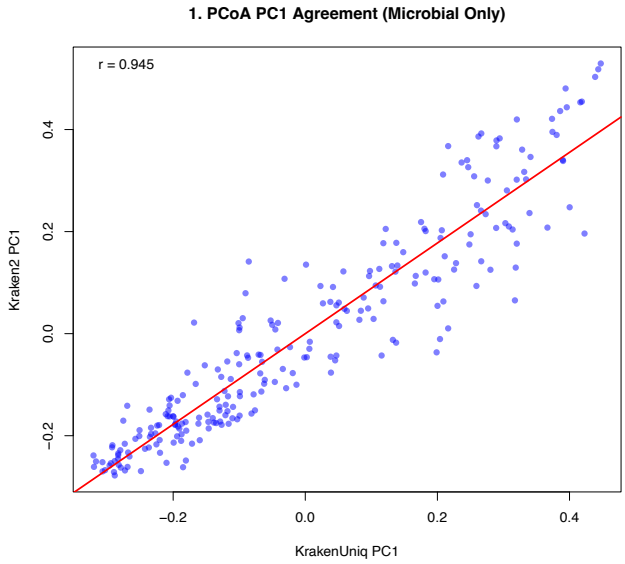
Beta-diversity was assessed using Bray-Curtis distances (computed on relative abundances) followed by Principal Coordinates Analysis (PCoA).

Result:
- **PC1 vs PC1 Spearman correlation r = 0.945 ($p < 0.001$)** between the two ordinations
- Global Procrustes Correlation r = 0.9003 ($p < 0.001$) for the full bivariate alignment

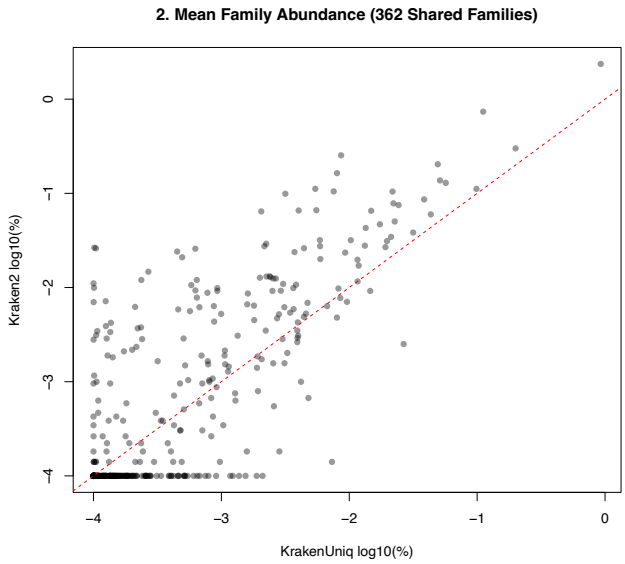
This very high correlation indicates that, while the tools may differ on individual family names, the global biological signal (the relationship between samples) is highly preserved.

Step 11: Visualization

Plot 1: PCoA PC1 Agreement (Main Result)
Displays the correlation between the first principal coordinates of both tools. A linear alignment with Spearman **r = 0.945** ($p < 0.001$) confirms that both tools capture the same dominant biological gradient.

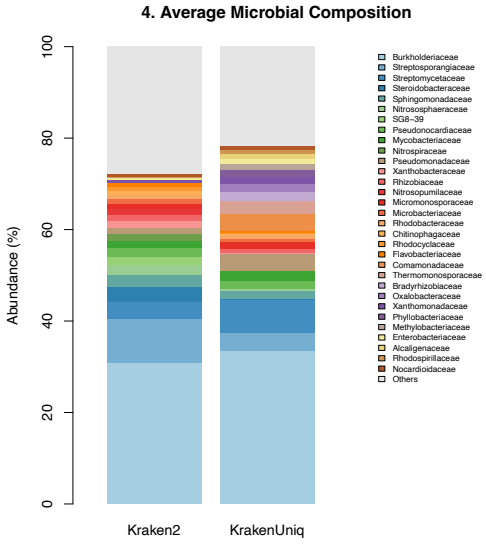


Plot 2: Mean Family Abundance (Log10 scale)
Comparison of mean percentage abundances across the 362 shared families.
Taxa align well along the 1:1 identity line, showing the quantification across databases for shared names.

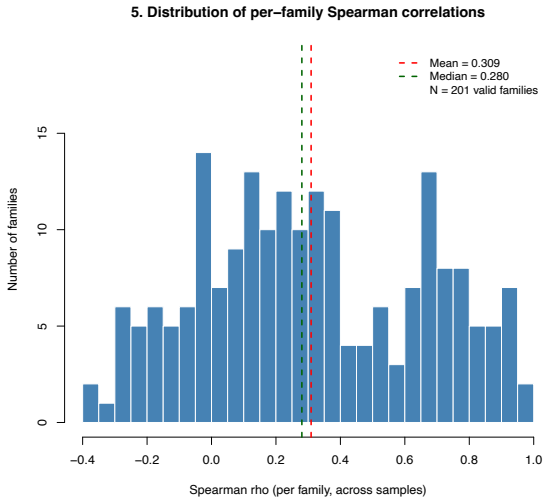


Plot 3: Shared Microbial Families (Venn Diagram)
Visualizes the nomenclature gap:
Kraken2 identifies 5,098 family names absent in KrakenUniq (mostly GTDB placeholders) vs **1,299 KrakenUniq-only family names**. Only **362 families overlap** by exact name, highlighting that divergence is primarily driven by database naming conventions.

Plot 4: Average Microbial Composition (Stacked Barplot)
Shows the Top 20 families normalized to 100% of the microbial fraction. Both tools identify *Burkholderiaceae* as the dominant taxon (~30% in Kraken2, ~33% in KrakenUniq), with highly consistent relative proportions for all major families.



Plot 5: Distribution of per-family Spearman correlations
Histogram of the 201 valid per-family Spearman rho values, with mean (red dashed) and median (green dashed) marked. The wide distribution (Q1 = 0.04, Median = 0.28, Q3 = 0.62, P95 = 0.89) clarifies that the moderate average masks two distinct regimes: **strong agreement on abundant families and stochastic disagreement on rare ones**.



Final Conclusions

The comparison reveals a "dual" reality:

- At the local taxonomic level, there is significant nomenclature divergence caused by database-specific conventions (GTDB placeholders vs. NCBI names). Of the 362 families sharing the same name across databases, fewer than half (201) actually had quantitative data in both tools, and only ~10% showed strong per-family agreement (rho > 0.8). The rare-taxa space is dominated by stochastic noise.
- At the global ecological level, the tools are highly consistent: the first principal coordinate of the two ordinations correlates at r = 0.945, and the full bivariate Procrustes analysis returns r = 0.90 ($p < 0.001$). The dominant biological signal, driven by the few abundant, well-classified families (Burkholderiaceae and other environmental taxa), is reproduced almost identically by the two pipelines.

For downstream analyses, both Kraken2 and KrakenUniq provide the same biological conclusions, despite their database-level differences in coverage and naming.